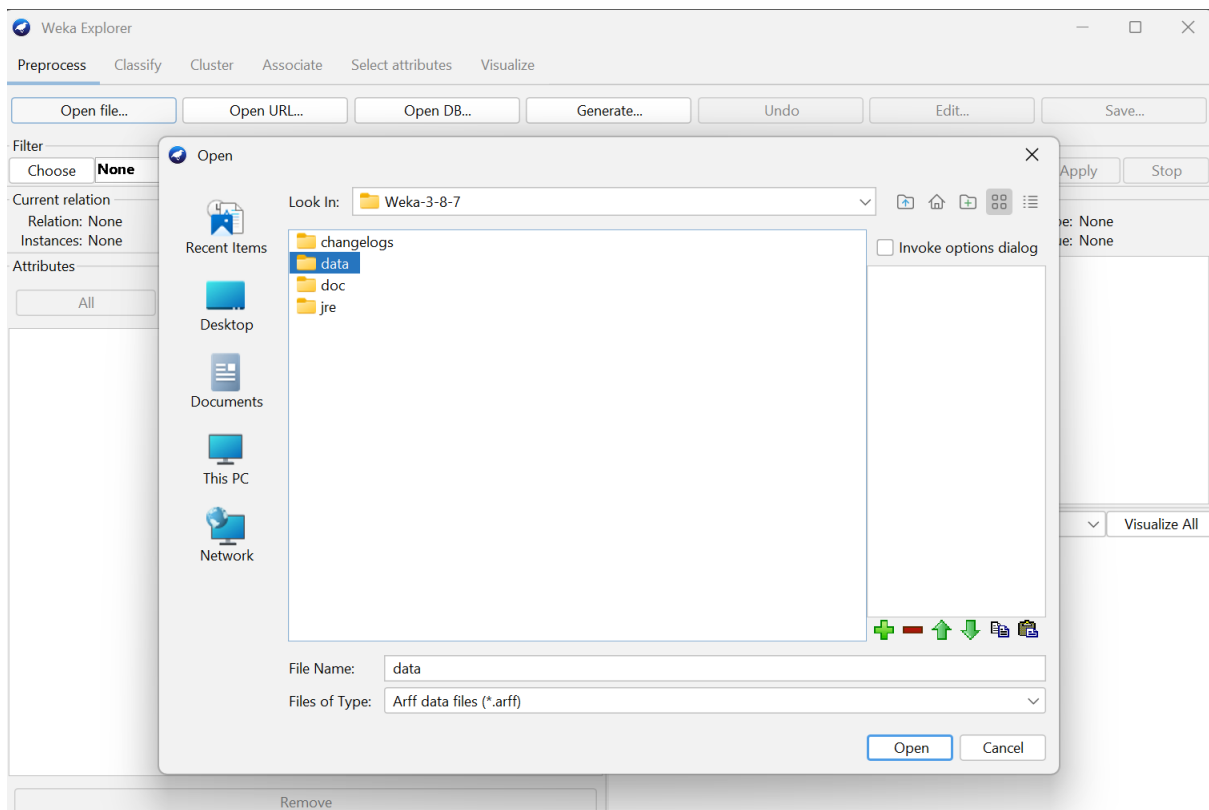
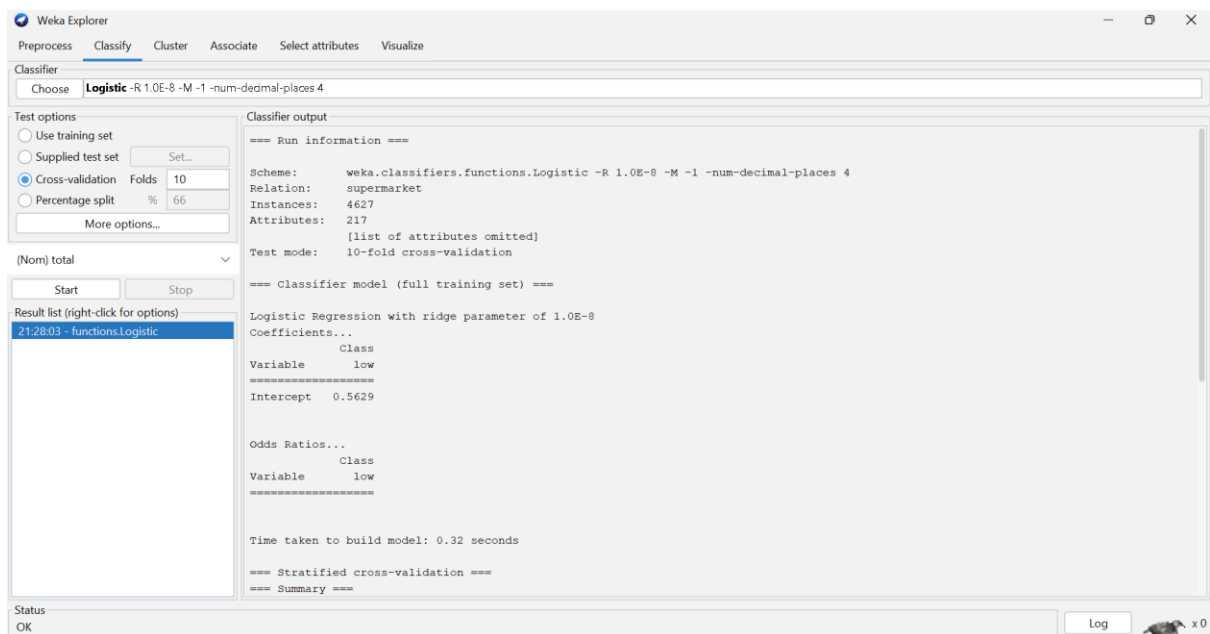


EXPERIMENT-29



A) Logistic Regression



Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **Logistic -R 1.0E-8 -M -1 -num-decimal-places 4**

Test options

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds **10**

☐ Percentage split % **66**

More options...

(Nom) total

Start Stop

Result list (right-click for options)

21:28:03 - functions.Logistic

Classifier output

Time taken to build model: 0.32 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	2948	63.713 %
Incorrectly Classified Instances	1679	36.287 %
Kappa statistic	0	
Mean absolute error	0.4624	
Root mean squared error	0.4808	
Relative absolute error	99.9961 %	
Root relative squared error	100 %	
Total Number of Instances	4627	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
1.000	1.000	0.637	1.000	0.778	?	0.499	0.637	low	
0.000	0.000	?	0.000	?	?	0.499	0.362	high	
Weighted Avg.	0.637	0.637	?	0.637	?	?	0.499	0.537	

=== Confusion Matrix ===

	a	b	<-- classified as
2948	0	1	a = low
1679	0	1	b = high

Status: OK

Log

B) Naïve Bayes Algorithm

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds **10**

☐ Percentage split % **66**

More options...

(Nom) total

Start Stop

Result list (right-click for options)

21:28:03 - functions.Logistic

21:30:24 - bayes.NaiveBayes

Classifier output

=== Run information ===

Scheme: weka.classifiers.bayes.NaiveBayes

Relation: supermarket

Instances: 4627

Attributes: 217

[list of attributes omitted]

Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

Naive Bayes Classifier

Attribute	Class	
	low	high
	(0.64)	(0.36)
department1		
t	556.0	493.0
[total]	556.0	493.0
department2		
t	60.0	73.0
[total]	60.0	73.0
department3		
t	53.0	39.0
[total]	53.0	39.0

Status: OK

Log

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation Folds
- ☐ Percentage split %
-

(Nom) total

Start Stop

Result list (right-click for options)

- 21:28:03 - functions.Logistic
- 21:30:24 - bayes.NaiveBayes**

Classifier output

department4		
t	48.0	38.0
[total]	48.0	38.0
department5		
t	88.0	89.0
[total]	88.0	89.0
department6		
t	1.0	3.0
[total]	1.0	3.0
department7		
t	40.0	29.0
[total]	40.0	29.0
department8		
t	1.0	1.0
[total]	1.0	1.0
department9		
t	51.0	33.0
[total]	51.0	33.0
grocery misc		
t	103.0	77.0
[total]	103.0	77.0
department11		

Status: OK

 x 0

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation Folds
- ☐ Percentage split %
-

(Nom) total

Start Stop

Result list (right-click for options)

- 21:28:03 - functions.Logistic
- 21:30:24 - bayes.NaiveBayes**

Classifier output

[total]		
	640.0	645.0
canned vegetables		
t	754.0	825.0
[total]	754.0	825.0
breakfast food		
t	930.0	934.0
[total]	930.0	934.0
cigs-tobacco pkts		
t	437.0	264.0
[total]	437.0	264.0
cigarette cartons		
t	18.0	21.0
[total]	18.0	21.0
cleaners-polishers		
t	588.0	676.0
[total]	588.0	676.0
coffee		
t	536.0	560.0
[total]	536.0	560.0
sauces-gravy-pkle		
t	1076.0	1127.0
[total]	1076.0	1127.0

Status: OK

 x 0

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options:
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds: **10**
☐ Percentage split %: **66**
 More options...

(Nom) total

Start Stop

Result list (right-click for options):
 21:28:03 - functions.Logistic
21:30:24 - bayes.NaiveBayes

Classifier output:

```

department209
t          1.0  1.0
[total]    1.0  1.0

department210
t          116.0 77.0
[total]    116.0 77.0

department211
t          110.0 99.0
[total]    110.0 99.0

department212
t          22.0  18.0
[total]    22.0  18.0

department213
t          13.0  11.0
[total]    13.0  11.0

department214
t          1.0   1.0
[total]    1.0   1.0

department215
t          1.0   1.0
[total]    1.0   1.0
  
```

Status: OK

Log

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options:
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds: **10**
☐ Percentage split %: **66**
 More options...

(Nom) total

Start Stop

Result list (right-click for options):
 21:28:03 - functions.Logistic
21:30:24 - bayes.NaiveBayes

Classifier output:

```

Time taken to build model: 0.05 seconds

=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances      2948           63.713 %
Incorrectly Classified Instances    1679           36.287 %
Kappa statistic                     0
Mean absolute error                 0.4624
Root mean squared error            0.4808
Relative absolute error             100 %
Root relative squared error         100 %
Total Number of Instances          4627

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
               1.000   1.000   0.637    1.000   0.778     ?      0.499   0.637   low
               0.000   0.000   ?        0.000   ?         ?      0.499   0.363   high
Weighted Avg.   0.637   0.637   ?        0.637   ?         ?      0.499   0.537

=== Confusion Matrix ===
  a    b  <-- classified as
2948  0 |  a = low
1679  0 |  b = high
  
```

Status: OK

Log

C) J48 Algorithm

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options:
☐ Use training set
☐ Supplied test set (Set...)
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
More options...

(Nom) total
Start Stop

Result list (right-click for options):
21:28:03 - functions.Logistic
21:30:24 - bayes.NaiveBayes
21:32:25 - trees.J48

Classifier output:

```
=== Run information ===  
  
Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2  
Relation:    supermarket  
Instances:   4627  
Attributes:  217  
              [list of attributes omitted]  
Test mode:   10-fold cross-validation  
  
=== Classifier model (full training set) ===  
  
J48 pruned tree  
-----  
: low (4627.0/1679.0)  
  
Number of Leaves :    1  
Size of the tree :    1  
  
Time taken to build model: 0.06 seconds  
  
=== Stratified cross-validation ===  
=== Summary ===  
  
Correctly Classified Instances      2948           63.713 %  
Incorrectly Classified Instances    1679           36.287 %  
Kappa statistic                     0  
Mean absolute error                 0.4624
```

Status: OK Log x 0

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options:
☐ Use training set
☐ Supplied test set (Set...)
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
More options...

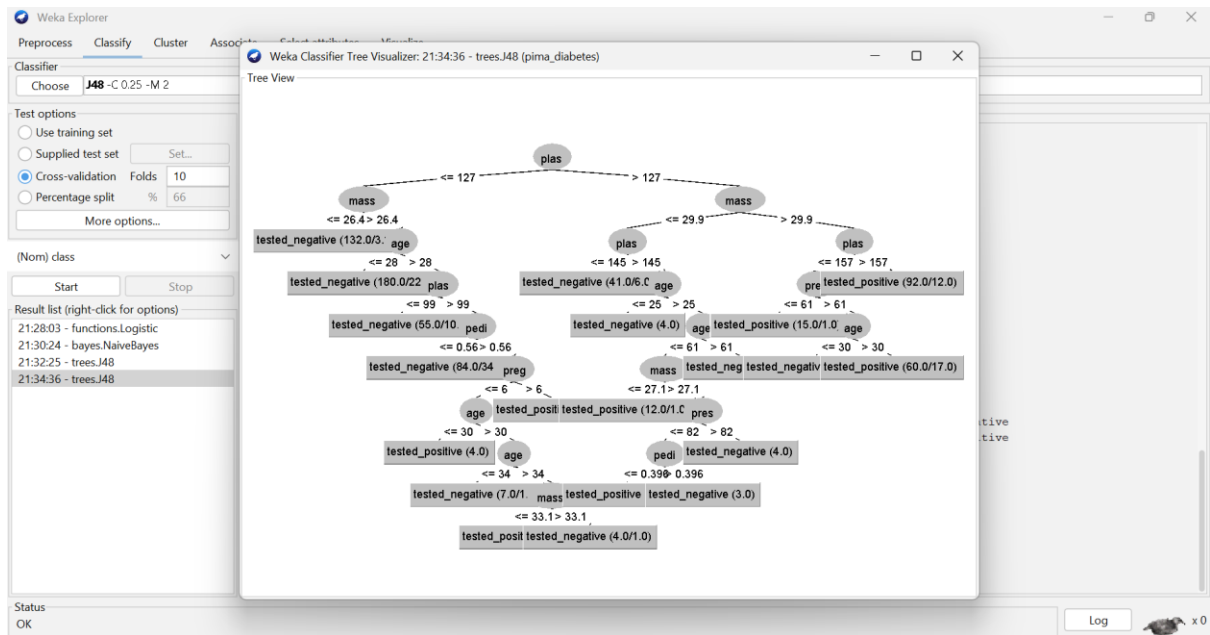
(Nom) total
Start Stop

Result list (right-click for options):
21:28:03 - functions.Logistic
21:30:24 - bayes.NaiveBayes
21:32:25 - trees.J48

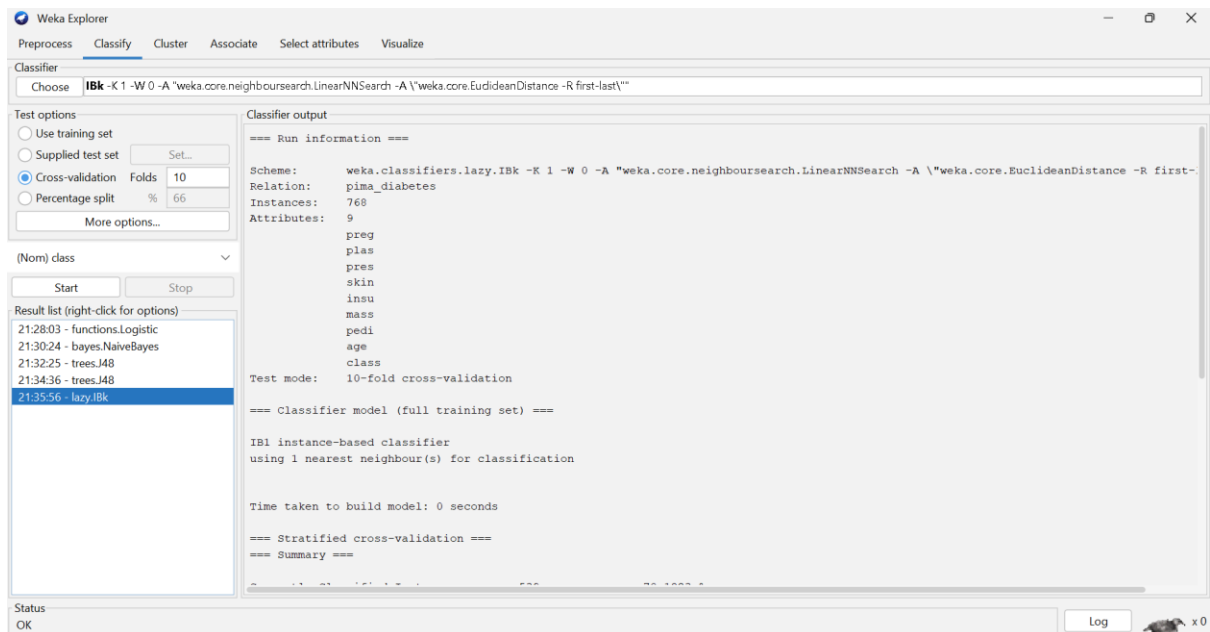
Classifier output:

```
Time taken to build model: 0.06 seconds  
  
=== Stratified cross-validation ===  
=== Summary ===  
  
Correctly Classified Instances      2948           63.713 %  
Incorrectly Classified Instances    1679           36.287 %  
Kappa statistic                     0  
Mean absolute error                 0.4624  
Root mean squared error             0.4808  
Relative absolute error             99.9961 %  
Root relative squared error         100 %  
Total Number of Instances          4627  
  
=== Detailed Accuracy By Class ===  
  
          TP Rate  FP Rate  Precision  Recall   F-Measure  MCC     ROC Area  PRC Area  Class  
1.000    1.000    0.637    1.000    0.778      ?     0.499    0.637    low  
0.000    0.000      ?         0.000      ?       ?     0.499    0.363    high  
Weighted Avg.   0.637    0.637      ?         0.637      ?       ?     0.499    0.537  
  
=== Confusion Matrix ===  
  
  a    b  <-- classified as  
2948  0 |  a = low  
1679  0 |  b = high
```

Status: OK Log x 0



D) K-Nearest Neighbor



Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **libk -K 1 -W 0 -A "weka.core.neighboursearch.LinearNNSearch -A \"weka.core.EuclideanDistance -R first-last\""**

Test options:
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
 More options...

(Nom) class: Start Stop

Result list (right-click for options):
 21:28:03 - functions.Logistic
 21:30:24 - bayes.NaiveBayes
 21:32:25 - trees.J48
 21:34:36 - trees.J48
21:35:36 - lazy.libk

Classifier output:

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
 === Summary ===

Correctly Classified Instances	539	70.1823 %
Incorrectly Classified Instances	229	29.8177 %
Kappa statistic	0.3304	
Mean absolute error	0.2988	
Root mean squared error	0.5453	
Relative absolute error	65.7327 %	
Root relative squared error	114.3977 %	
Total Number of Instances	768	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Weighted Avg.	0.794	0.470	0.759	0.794	0.776	0.331	0.650	0.732	tested_negative
	0.530	0.206	0.580	0.530	0.554	0.331	0.650	0.469	tested_positive
	0.702	0.378	0.696	0.702	0.698	0.331	0.650	0.640	

=== Confusion Matrix ===

```

a  b  <-- classified as
397 103 | a = tested_negative
126 142 | b = tested_positive

```

Status: OK

E) SMO Algorithm

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **SMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-8 -M -1 -num-decimal-places 4"**

Test options:
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
 More options...

(Nom) class: Start Stop

Result list (right-click for options):
 21:28:03 - functions.Logistic
 21:30:24 - bayes.NaiveBayes
 21:32:25 - trees.J48
 21:34:36 - trees.J48
 21:35:36 - lazy.libk
21:36:48 - functions.SMO

Classifier output:

=== Run information ===

```

Scheme:      weka.classifiers.functions.SMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.Poly
Relation:    pima_diabetes
Instances:   768
Attributes:  9
  preg
  plas
  pres
  skin
  insu
  mass
  pedi
  age
  class
Test mode:   10-fold cross-validation

```

=== Classifier model (full training set) ===

```

SMO
Kernel used:
  Linear Kernel: K(x,y) = <x,y>
Classifier for classes: tested_negative, tested_positive
BinarySMO

```

Status: OK

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Classifier

ChooseSMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-8 -M -1 -num-decimal-places 4"

Test options

☐ Use training set

☐ Supplied test set

Set...

☒ Cross-validation

Folds10

☐ Percentage split

%66

More options...

(Nom) class

Start

Stop

Result list (right-click for options)

21:28:03 - functions.Logistic

21:30:24 - bayes.NaiveBayes

21:32:25 - trees.J48

21:34:36 - trees.J48

21:35:56 - lazy.IBk

21:36:48 - functions.SMO

Classifier output

Time taken to build model: 0.03 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	594	77.3438 %
Incorrectly Classified Instances	174	22.6563 %
Kappa statistic	0.4682	
Mean absolute error	0.2266	
Root mean squared error	0.476	
Relative absolute error	49.848 %	
Root relative squared error	99.862 %	
Total Number of Instances	768	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.898	0.459	0.785	0.898	0.838	0.480	0.720	0.771	tested_negative
	0.541	0.102	0.740	0.541	0.625	0.480	0.720	0.560	tested_positive
Weighted Avg.	0.773	0.334	0.769	0.773	0.763	0.480	0.720	0.698	

=== Confusion Matrix ===

a

b

<-- classified as

449

51

|

a = tested_negative

123

145

|

b = tested_positive

Status

OK

Log

x 0